Assignment 1

Lab on - January 6, 2026

The goal of this assignment is to understand software profiling using **GNU gprof**. You will implement two sorting algorithms with different time complexities namely **Bubble Sort** ($O(n^2)$) and **Quick Sort** ($O(n \log n)$) and analyze their runtime performance across varying input sizes.

Step 1: Implementation

- Create a C program named `sorting_profile.c` that accepts two command-line arguments
  1. 'n' for the input size.
  2. one of the sorting algorithms 'bubble' or 'quick'
- The program should allocate an array of size n with random integers. Implement `bubble_sort()` and `quick_sort()` as separate functions.

Step 2: Compiling and Profiling

- Compile your code using `gcc` with `-pg` flag. This instruments the code for profiling.
- Disable any compiler optimizations (use `-O0`) to ensure the profiler tracks the performance without the compiler inlining functions or unrolling loops etc.

Step 3: Generate Data

- Run both sorting algorithms with 'n' ranging from 5000 to 50,000 (increments of 5000).
- After each run, execute the `gprof` to analyze the generated `gmon.out` file. As an example, the command will look like: `gprof -b sorting_profile gmon.out`
- Extract the time spent in function for the specific sorting function (self-seconds).

Step 4: Visualization

- Plot the data (input size vs time) for both algorithms on the same graph.

Step 5: Upload

Upload the following as a single *.zip* file with your roll number as the file name. The zip file should contain the following:

- The C code containing the profiler code along with the sorting algorithms
- The plotting code along with the plot PDF file

Tutorial on plotting data

Assume, you have the following data as *data.txt* where the first column represent months and the next two columns represent attendance in lab and attendance in theroy class respectively.

```
1, 10, 5
2, 25, 12
3, 15, 8
4, 40, 20
5, 30, 15
6, 55, 30
7, 45, 25
```

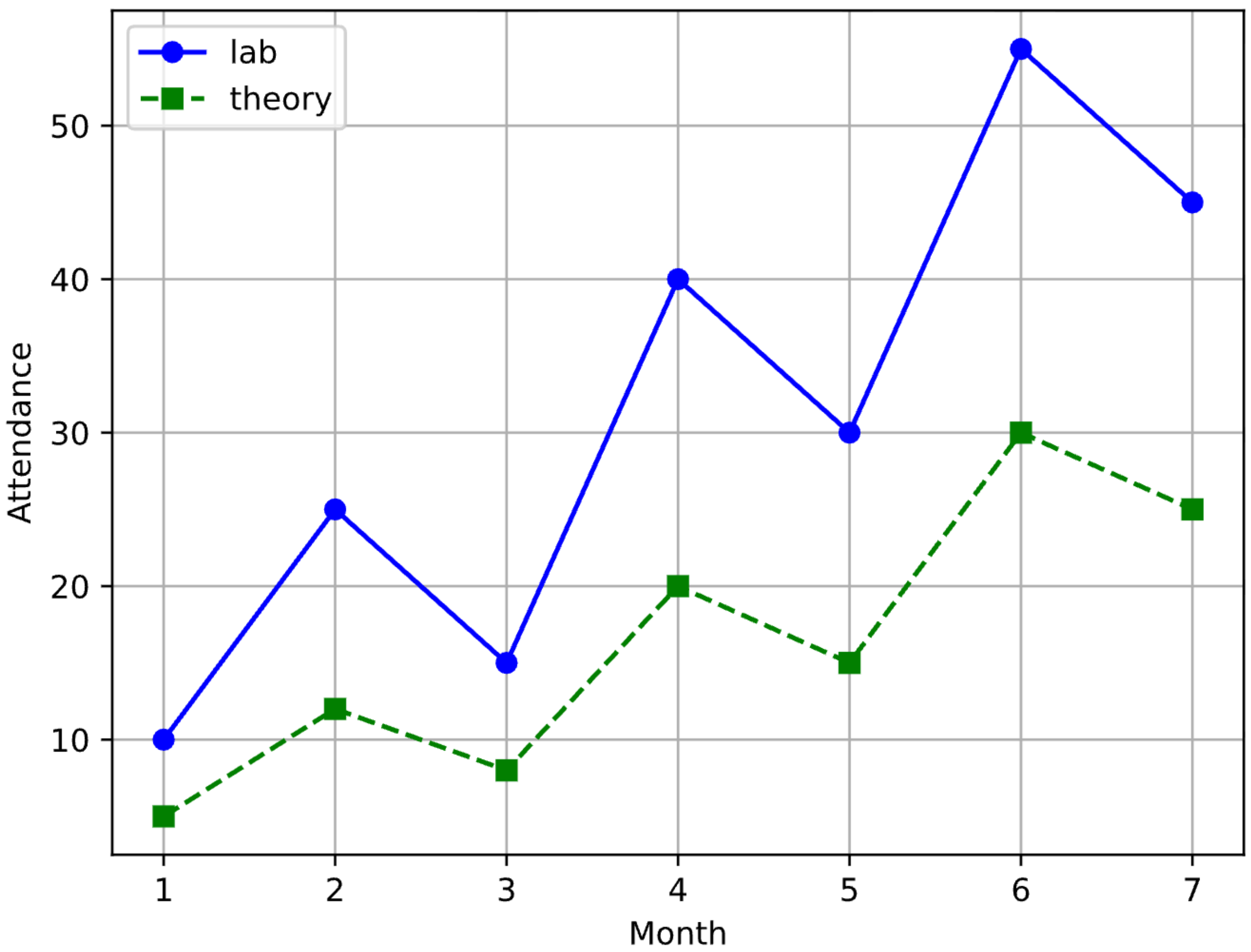
We will a Python library called `matplotlib` to plot this data.

Use the following code and save as *plot.py*.

```
import matplotlib.pyplot as plt

months = []
lab_attendance = []
theory_attendance = []

# First, we will read the text file
with open('data.txt', 'r') as f:
    for line in f:
        parts = line.split(',')
        months.append(float(parts[0])) # first columns
        lab_attendance.append(float(parts[1])) # second column
        theory_attendance.append(float(parts[2])) # third column

# Create the lab attendance plot
plt.plot(months, lab_attendance, label="lab", marker='o', linestyle='-',
color='blue')
# Create the theory attendance plot
plt.plot(months, theory_attendance, label="theory", marker='s', linestyle='-
-', color='green')

# Set the x and y labels for the plot
```

```
plt.xlabel("Month")
plt.ylabel("Attendance")

plt.grid(True)
plt.legend()
plt.savefig("attendance-plot.pdf") # Save the attendance plot as a PDF file
plt.close()
```

Plot the data using the command `python plot.py`. This should create a new PDF file with the plot which look as below.



You should use the above tutorial to create a PDF file for the plot for running times of bubble and quick sort.